

The ESM-1b Transformer Network

What is the ESM-1b Transformer Network?

RESEARCH ARTICLE | BIOLOGICAL SCIENCES | 



Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences

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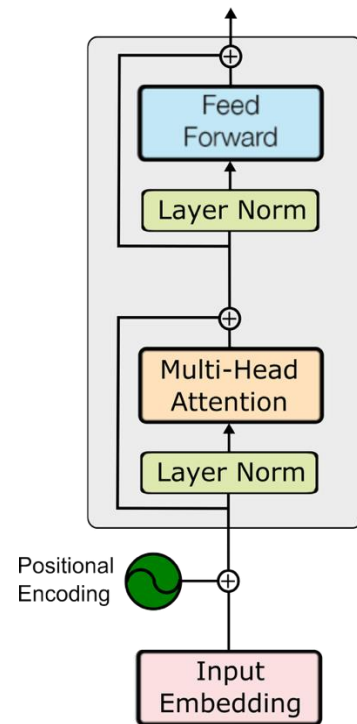
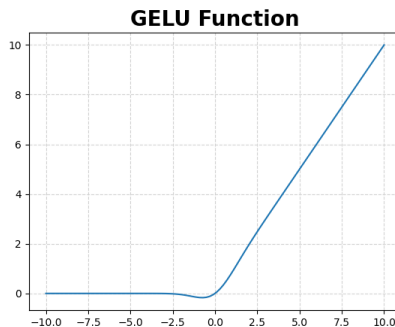
- Open-Source protein language model by the Facebook AI research team
- Architecture: Transformer Network Encoder
- Training for Masked Language Modeling (MLM)

ESM-1b Architecture – Encoder

- Each encoder layer
 - Dimension of representations $d = 1280$
 - Number of attention heads $h = 20$
 - Maximum sequence length: 1024
 - Hidden dimension of FFN: 5120 ($4 \cdot d$)
- 33 encoder layers: 652.4 M learnable parameters
- Layer Normalization before Attention and FFN block
- Activation of Feed Forward Neural Network: GELU instead of ReLU

$$\text{GELU}(x) = x \cdot \Phi(x)$$

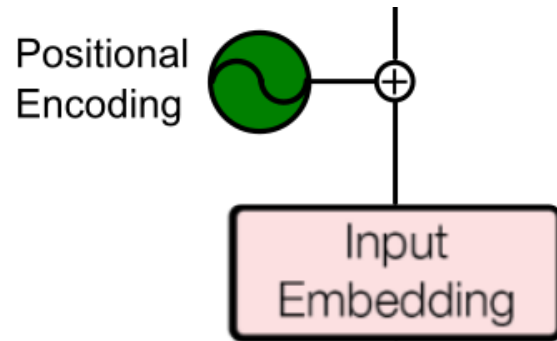
Φ = cumulative distribution function (CDF) of the standard normal distribution



ESM-1b Architecture – Input Embeddings

- Token embeddings are learned with a dimension of $d = 1280$
- Sinusoidal positional encodings were compared to learned positional encodings:
 - Learned positional embeddings led to better results
- How can we learn token (and positional) embeddings?

$$\begin{pmatrix} w_{1,1} & \cdots & w_{23,1} \\ \vdots & \ddots & \vdots \\ w_{1,1280} & \cdots & w_{23,1280} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} = \begin{pmatrix} w_{2,1} \\ w_{2,2} \\ \vdots \\ w_{2,1280} \end{pmatrix}$$



- For positional embeddings (max. sequence length of 1024), a matrix of dimension 1024×1280 is learned.

ESM-1b Training

- Training task: Masked Language Modeling (MLM)

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 - Randomly select 15% of the input tokens that shall be predicted: Of these:
 - 80% are replaced with the mask token
 - 10% are placed with a random amino acid
 - 10% are not changed
- Tied input/output embeddings
 - The transpose of the learnable input embeddings $W^T \in \mathbb{R}^{1280 \times 23}$ is used to map the output embeddings $\vec{h}_i$ (at position i) to token predictions (logits)

$$W^T \cdot \vec{h}_i = \begin{pmatrix} \vec{w}_1^T \\ \vdots \\ \vec{w}_{23}^T \end{pmatrix} \cdot \begin{pmatrix} h_{i,1} \\ \vdots \\ h_{i,1280} \end{pmatrix} = (\vec{w}_1^T \cdot h_i, \dots, \vec{w}_{23}^T \cdot h_i) \in \mathbb{R}^{23}$$

- Advantages: Fewer parameters and better generalization
- Training dataset: UniRef50 dataset with $\sim 30M$ protein sequences

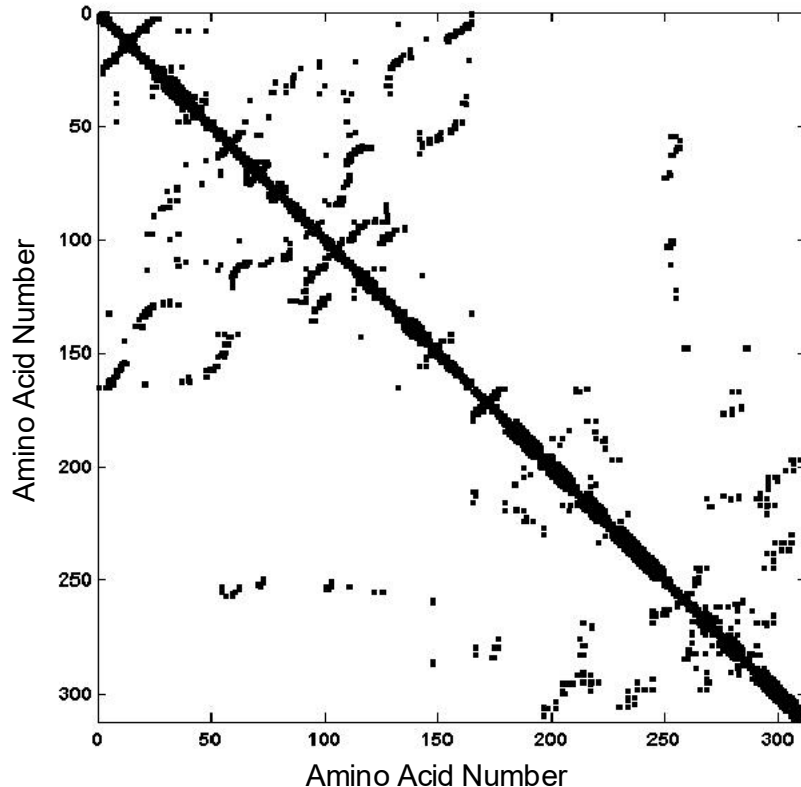
ESM-1b Training (2)

- Training time: ~20 days of 64 NVIDIA V100 GPUs
 - Each epoch took 8.5 hours
 - 56 epochs in total
- Optimizer: Adam ($lr = 10^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$)
- Dynamic batching for faster training
 - All sequences are truncated to ≤ 1024 tokens
 - Sequences are grouped into batches based on their length
 - Shorter sequences $\rightarrow$ more per batch; longer ones $\rightarrow$ fewer
 - Each batch contains a set of variable-length sequences (padded to the longest in the batch), but the token count per batch remains roughly constant



Component	Standard Transformer	ESM-1b
Dropout in attention layers	Yes	No
Dropout in FFN layers	Yes	No
Token-level dropout	No (applied to comb. word and position emb.)	Yes (token-level dropout for masked tokens)

Applications – Contact Map Prediction



- Protein contact maps
 - Black dots indicate which amino acids are close to each other in the protein 3D structure
- Predicting Contact Maps using amino acid representations shows learned structural information
 - Binary prediction problem:
 - Let h_i and h_j be the representations of amino acids i and j
 - $f(h_i, h_j) = \text{sigmoid}(\langle P \cdot h_i, Q \cdot h_j \rangle + b) \in [0,1]$

Applications – Function Prediction

- ESM-1b vectors (whole protein representations) have been successfully used for all kinds of protein function predictions:
 - Enzyme functions
 - Substrates for transport proteins
 - Protein-Protein interactions
 - Drugs inhibiting proteins
 - ...
- To solve these prediction tasks, the fixed ESM-1b vectors $\vec{e}$ (element-wise mean of all amino acid representations) can be used as the input of new prediction models

$$f(\vec{e}) = y$$

How to use the model

```
import torch
import esm # install via pip install fair-esm==0.4.0
model, alphabet = torch.hub.load("facebookresearch/esm:v0.4.0", "esm1b_t33_650M_UR50S")
batch_converter = alphabet.get_batch_converter()
model.eval(); # disables dropout for deterministic results
```

[1] ✓ 7.1s

Python

```
protein = "MKYFPLFPTLVFAARVVAFFPAYASLAGLSQQLDAIIPTLEAREPLPPGLENSAKLVNDEAHPWKPLRPGDIRGPCGLNTLASHGYLPRNGVATPVQIINAVQEGLN"
print(len(protein))
#Generate per-sequence representations via averaging
batch_labels, batch_strs, batch_tokens = batch_converter([("protein1", protein)])
with torch.no_grad():
    results = model(batch_tokens, repr_layers=[33], return_contacts=True)

#token 0 is a special token, so the first amino acid is token 1
token_representations = results["representations"][33]
sequence_representations = token_representations[0, 1:len(protein) + 1].mean(0)
sequence_representations, sequence_representations.shape
```

[2] ✓ 1.2s

Python

... 371

```
... (tensor([ 0.0207,  0.1680,  0.0377, ...,  0.2453, -0.2084,  0.0769]),
     torch.Size([1280]))
```